THRIFTKART / SEMESTER III

Estimation Sheet

Three-point estimates, PERT calculations and capacity conversion

Formula	TE = (O + 4M + P) / 6
Unit	Productive person-days
Team capacity	12 hours/day = 2 productive days/day
Weekly capacity	10 productive person-days/week
Total TE	24.50 productive person-days
Caution	Estimate, not a promise

1. Case-Provided Estimates

Work package	O	M	P	Numerator	TE
Listing & search	4	6	12	4 + 24 + 12 = 40	40/6 = 6.67
Chat between buyer and seller	3	5	10	3 + 20 + 10 = 33	33/6 = 5.50
Payments	3	4	9	3 + 16 + 9 = 28	28/6 = 4.67
Admin moderation	2	3	6	2 + 12 + 6 = 20	20/6 = 3.33
Testing	3	4	7	3 + 16 + 7 = 26	26/6 = 4.33

2. Total Effort

Total TE = 6.67 + 5.50 + 4.67 + 3.33 + 4.33 = 24.50 productive person-days.

3. Capacity Conversion

Step	Calculation	Result
Daily team hours	4 students × 3 hours/day	12 hours/day
Productive day conversion	12 ÷ 6 hours	2 productive days/day
Weekly capacity	2 × 5 weekdays	10 productive days/week
Serial calendar conversion	24.50 ÷ 10	2.45 weeks
Critical-path conversion	21.17 ÷ 10	2.12 weeks

Interpretation

The 2.45-week figure is the serial conversion of total effort. The 2.12-week critical-path figure is the network duration under the stated parallel branch. Neither is a promise because student availability, coordination and rework are sources of uncertainty.

4. Why PERT Is an Estimate

- Optimistic, most-likely and pessimistic values capture uncertainty in the same task.
- The expected-time formula weights the most-likely estimate four times, but the result remains sensitive to assumptions and execution conditions.
- The case also removes all team capacity in weeks 9 and 10, so the practical schedule must preserve early completion rather than rely on those weeks.